Computational Neurodynamics

Exercise 1

Programming exercises for Topics 2–3: Numerical integration and neuron models

1. **Numerical integration**. The systems of ordinary differential equations (ODEs) that describe spiking neurons are usually impossible to solve analytically – at least in the complex settings we are usually interested in. In those cases, we must resort to numerical integration methods.

   The simplest method of numerical integration is the Euler method. To integrate a continuous function $y(t)$, we discretise time in steps of width δt.

   $$y'(t) = f(t, y)$$
   $$y(t + \delta t) = y(t) + \delta t\, f(t, y)$$

   Where $y'(t)$ represents the time derivative of y at time t.

   (a) Write a Python script that implements the Euler method and solve the differential equation

   $$\frac{dy}{dt} = y$$

   Solve this equation for different values of δt and compare it against the analytical solution, $y(t) = \exp(t)$.

   (b) The Euler method can also be applied to second-order ordinary differential equations. For example, a damped oscillator can be described by the following second-order ODE:

   $$m\, y''(t) = -cy' - ky$$

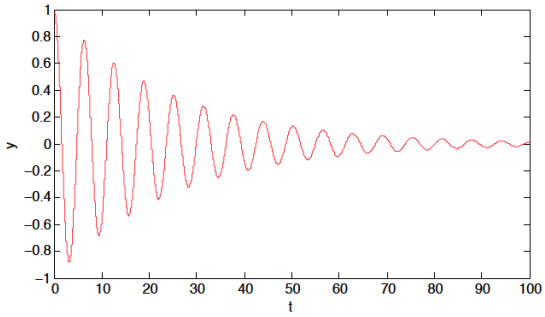
   where m is the mass, c is the damping coefficient, k is the spring constant, and y' and y'' denote the first and second time derivatives of y, respectively. This can be solved with the Euler method by defining another variable

   $$p(t) = y'(t)$$

   Thus turning a second-order ODE into two coupled first-order ODEs (one for $y(t)$ and one for $p(t)$). Write a Python script to simulate a damped oscillator system using the Euler method. As initial conditions, let

   $$y(0) = 1$$
   $$p(0) = 0$$

   Taking $m = 1$, $c = 0.1$, and $k = 1$, solve the damped oscillator with the Euler method and reproduce the plot below. If you don't see the results you would expect, try reducing the integration step size.



(c) Implement the 4th-order Runge-Kutta (RK4) method and use it to solve the damped oscillator system above.

2. **The double pendulum**. Let's look at some more interesting (but still non-neural) ODEs and evaluate the performance of the two numerical integration methods we have covered so far. Let us focus on the compound *double pendulum*, a physical system comprised by two connected rods (imagine one pendulum hanging from the end of another pendulum). The double pendulum is given by two angular variables θ_i and two momentum variables p_i, governed by the coupled ODEs:

$$\theta_1' = \frac{6}{ml^2}\,\frac{2p_1 - 3\cos(\theta_1-\theta_2)\,p_2}{16 - 9\cos^2(\theta_1-\theta_2)}$$

$$\theta_2' = \frac{6}{ml^2}\,\frac{8p_2 - 3\cos(\theta_1-\theta_2)\,p_1}{16 - 9\cos^2(\theta_1-\theta_2)}$$

$$p_1' = -\frac{1}{2}ml^2\left(\theta_1'\theta_2'\sin(\theta_1-\theta_2) + 3\frac{g}{l}\sin\theta_1\right)$$

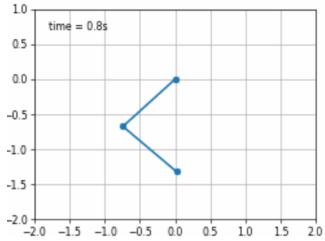
$$p_2' = -\frac{1}{2}ml^2\left(-\theta_1'\theta_2'\sin(\theta_1-\theta_2) + \frac{g}{l}\sin\theta_2\right)$$

Where m represents the mass of each rod, l represents their length, and g is the gravitational acceleration. Suggested values for an Earth-based pendulum are $l = m = 1$ and $g = 9.8$. For visualisation purposes, you can transform these polar coordinates into Cartesian coordinates:

$$x_1 = l\sin\theta_1 \qquad y_1 = -l\cos\theta_1$$

$$x_2 = x_1 + l\sin\theta_2 \qquad y_2 = y_1 - l\cos\theta_2$$

And display them in a plot like this:



(a) Simulate the compound double pendulum with the Euler and RK4 methods, starting from the initial conditions $\theta_1 = 120°$, $\theta_2 = -10°$, and $p_1 = p_2 = 0$. Use a relatively small integration time-step (e.g. $\delta t = 0.01\,\text{s}$) and simulate for a relatively short period of time (e.g. $T = 5\,\text{s}$).

(Note that, even then, this system is so unstable that it will quickly spiral out of control – and in fact it will stop obeying the conservation of energy due to the approximations in both Euler and RK4. If you think this doesn't make sense, you're probably right. Head on to your favourite online video platform and check out some videos for the compound double pendulum to convince yourself that double pendulums indeed don't make sense anyway.)

(b) Write a function to quantify the difference between two trajectories of a dynamical system. One of many suitable functions could be the *mean absolute error* (MAE), which is simply the mean absolute difference between both trajectories at each timestep. Let's first try this out on the damped oscillator from Question 1: Given two simulations of the damped oscillator (say, one with Euler and one with RK4), plot how their MAE evolves over simulation time. How do you interpret this curve?

(c) Repeat Question 2b for the double pendulum. This curve differs in an important sense from its analogue for the damped oscillator. Why?

3. **Simulating the Hodgkin-Huxley model**.

(a) Using the equations and constants on the slides from Topic 2 (Neurons), use the Euler method to simulate a single neuron using the Hodgkin-Huxley model. Verify that your code reproduces the plot of the Hodgkin-Huxley model in the notes, given a constant dendritic current of $I = 10$. Use initial values of 0 for v, m, n, and h, and an integration step of $\delta t = 0.01$.

(b) Reimplement your code from part (a) using the Runge-Kutta 4 method.

Note: in this form of the model, voltages are measured relative to the resting potential (approximately $-65\,\text{mV}$), so rest corresponds to $v \approx 0$.

Written questions for Topics 2–3

4. **Anatomy of a neuron**.

   (a) Draw a schematic diagram of a typical cortical neuron. Label three important parts of the neuron and briefly describe their role in the neuron's signalling process.

   (b) "The two main morphological types of neurons are excitatory and inhibitory". Is this statement true or false? Why?

5. **Action potentials in the Hodgkin-Huxley model**. The Hodgkin-Huxley model is given by the differential equation

$$C\frac{dv}{dt} = -\left(I_{\mathrm{Na}} + I_{\mathrm{K}} + I_{\mathrm{L}}\right) + I$$

   Where the I terms refer to currents related to sodium, potassium, leakage, and external input, respectively.

   (a) When fed a constant external current, the Hodgkin-Huxley model produces a steady sequence of action potentials. Draw a typical time course of an action potential and label all relevant parts.

   (b) Describe the neuronal process that generates action potentials when the neuron is fed a constant external current, and relate each part of the process (where applicable) to the components of the Hodgkin-Huxley model.

   (c) When the neuron's membrane potential is v, the leakage current is given by $I_{\mathrm{L}} = g_{\mathrm{L}}(v - E_{\mathrm{L}})$, for some value of scalar parameters $g_{\mathrm{L}}, E_{\mathrm{L}}$. What is the intuitive meaning of the parameter E_{L}? Justify your answer mathematically.

6. **Numerical integration algorithms**.

   (a) In the context of numerical integration, what is an initial value problem? In what context is it used in computational neuroscience?

   (b) Name and describe two numerical integration algorithms, defining all necessary equations and quantities involved. Briefly summarise the advantages of each algorithm over the other.